A Dual-Stage Hybrid Pipeline for Aerial Landmine Detection in LWIR Thermal Imagery

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***Abstract*—Thermal imaging from unmanned aerial systems (UAS) offers a stand-off modality for humanitarian demining, yet buried landmine signatures in long-wave infrared (LWIR) bands are easily confounded by environmental thermal clutter such as sun-heated rocks, vegetation, and soil temperature gradients. This paper presents a dual-stage hybrid pipeline that couples a YOLO26 one-stage detector with a Random Forest (RF) classifier operating on ten handcrafted thermal and geometric features. The first stage achieves 95.78% recall on the 1,320-image AMLID test partition. The second stage filters false positives using features that encode shape compactness, radial thermal symmetry, edge structure, and frequency-domain texture, reducing false alarms by 31.7% (865 to 591) while maintaining 90.86% recall. The combined system yields a peak overall mAP@50 of 91.9%, and the AP-Metal class achieved 84.6% mAP@50. The AMLID paper reports 19.3% Precision@50 for the same class, but these metrics are not directly comparable. The optimized model compresses to 20.3 MB under FP16 quantization and sustains 204 FPS inference, supporting low-latency edge inference.**

I. Introduction

Landmines remain a persistent humanitarian threat across more than 60 countries, with an estimated 110 million buried devices causing thousands of casualties annually [1]. Airborne thermal imaging provides a stand-off detection modality, as buried mines alter the thermal conductivity and heat capacity of the overlying soil layer, producing a measurable surface temperature differential in LWIR bands [2], [3], [4]. The AMLID (Adaptive Multispectral Landmine Identification Dataset) corpus [5] provides a standardized benchmark of 12,078 LWIR frames annotated across four mine classes (AT-Plastic, AT-Metal, AP-Plastic, AP-Metal), enabling reproducible comparison of detection architectures.

Deep convolutional detectors such as YOLO [6] achieve spatial localization on this task, yet they remain vulnerable to thermal environmental clutter. Sun-heated rocks, compacted soil patches, and metallic debris produce thermal signatures whose spatial statistics partially overlap with those of mine simulants, leading to elevated false-positive rates. The published AMLID baseline [5] reports a YOLOv8 [7] model achieving 86.8% mean average precision (mAP@50), with per-class precision for the AP-Metal category falling to 19.3% — a level insufficient for operational deployment.

This work develops a dual-stage architecture in which a YOLO26 Small network proposes candidate regions at high recall, and a Random Forest classifier subsequently verifies each proposal against ten handcrafted features. Handcrafted features grounded in the known physics of buried mine signatures — circular shape, uniform thermal distribution, radial heat diffusion symmetry, and engineered edge geometry — act as a geometric and thermodynamic gate against natural clutter. The contributions are:

1) A YOLO26 detector evaluated on the AMLID test partition achieving 91.9% mAP@50, with per-class mAP@50 of 99.1% (AT-Plastic), 97.4% (AT-Metal), 86.7% (AP-Plastic), and 84.6% (AP-Metal).
2) A geometric-physics feature set comprising ten interpretable measurements — area, circularity, thermal contrast, aspect ratio, radial thermal symmetry, Laplacian-of-Gaussian zero-crossing count, rotation-invariant local binary patterns, DCT high-frequency energy ratio, wavelet approximation energy, and internal hole count — each designed to encode a distinct physical property of buried mine signatures.
3) A soft-voting ensemble that fuses YOLO confidence with RF probability, reducing false positives by 31.7% (865 to 591) and achieving a system-level F1 score of 93.27%.
4) A compact model at 20.3 MB running at 204 FPS on a Tesla T4 GPU, supporting low-latency UAS-mounted inference.

II. Related Work

A. Deep Object Detectors for Mine Detection

One-stage detectors, particularly the YOLO family [6], [8], have been applied to landmine detection in LWIR imagery owing to their speed-accuracy trade-off. Gallagher and Oughton [5] trained YOLOv8 on the AMLID dataset and

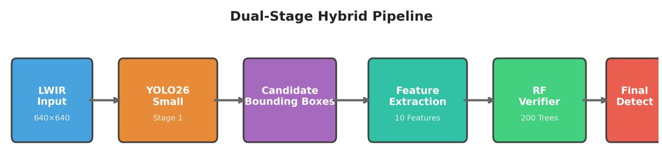

Fig. 1. Overview of the proposed dual-stage hybrid detection pipeline. YOLO26 proposes candidate regions at high recall; the Random Forest verifier filters false positives using ten handcrafted thermal and geometric features.

reported an overall mAP@50 of 86.8%, but observed severe degradation on the AP-Metal class (19.3% precision), attributing the failure to the small spatial footprint and low thermal contrast of anti-personnel mines. Beyond the YOLO family, single-shot detectors such as SSD [9] offer comparable speed with multi-scale feature maps, while transformer-based detectors like DETR [10] eliminate hand-designed components such as anchor boxes and non-maximum suppression. More recent Ultralytics models (YOLOv8 [7], YOLO11, YOLO26) continue to improve the speed-accuracy frontier, though their application to thermal mine detection remains relatively unexplored. These results suggest that purely data-driven detectors lack the inductive bias needed to reject non-mine thermal anomalies.

B. Classical Feature-Based Detectors

Prior to the dominance of deep learning, landmine detection relied on statistical classifiers operating on engineered features. Lopez et al. [3] modeled thermal contrast of buried mines using heat-transfer analysis, while Cremer et al. [4] applied Hough transform and morphological filters to extract circularity and shape features from thermal IR images. Random Forests (RF) [11] have been widely adopted for classification in computer vision due to their robustness to high-dimensional feature spaces and built-in feature importance ranking. These methods offer interpretability but lack the spatial context that deep features capture.

C. Hybrid and UAV-Based Detection

UAV-based platforms have gained attention for large-area minefield survey. Bajić and Potočnik [12] applied YOLOv5 to thermal UAV imagery for unexploded ordnance (UXO) detection across eleven classes, achieving high per-class detection rates. Qiu et al. [13] proposed a detection-driven multispectral fusion framework using YOLOv5 for UAV-borne scatterable landmine detection, demonstrating that combining spectral bands improves recall under occlusion. Hybrid detector-verifier pipelines have also been explored in medical imaging and industrial inspection, where a deep detector proposes regions and a shallow classifier verifies them. This work applies this paradigm to LWIR thermal imagery, using a feature set designed for the thermal and geometric properties of buried mines.

III. Methodology

A. Dataset and Preprocessing

The AMLID corpus [5] comprises 12,078 LWIR thermal images collected across multiple months, times of day (post-sunrise, noon, pre-sunset), and UAS elevations (5–20 m: 5 m, 10 m, 15 m, 20 m). Annotations follow the Pascal VOC format with four classes: AT-Plastic, AT-Metal, AP-Plastic, and AP-Metal. The hierarchical directory structure was flattened into unique filenames encoding capture parameters (e.g., `Jan_Jan_Afternoon_10_lwir_3.jpg`). Following the AMLID benchmark protocol [5], the official test partition of 1,320 images (14,905 ground-truth instances) was used for all reported evaluations. The remaining 10,758 images were split into 9,682 training and 1,076 validation images (approximately 90/10) for hyperparameter selection and early stopping. All images were resized to 640×640 pixels for YOLO training while preserving the original aspect ratio with letterbox padding.

B. Stage 1 — YOLO26 One-Stage Detector

YOLO26 Small [8] was selected as the base detector for its balance of inference speed and detection accuracy. YOLO26 is the latest release in the YOLO series at the time of this work, incorporating architectural improvements over prior versions. A systematic comparison across YOLO versions (v8, v11, etc.) and alternative architectures such as RT-DETR is left as future work, as the primary focus here is the hybrid pipeline design rather than detector selection. Training was performed on a single Tesla T4 GPU (16 GB VRAM) via Kaggle with the following configuration:

- **Pretrained weights:** YOLO26 Small (`yolo26s.pt`)
- **Input resolution:** 640 $\times$ 640 pixels
- **Batch size:** 32
- **Epochs:** 50 (early stopping patience of 15, not triggered; best checkpoint at epoch 49)
- **Optimizer:** AdamW (weights stripped post-training)
- **Workers:** 4

Training loss and mAP@50 curves across all 50 epochs are shown in Fig. 3.

Training ran for the full 50 epochs; the best checkpoint was recorded at epoch 49 with mAP@50 = 0.918, precision = 0.907, and recall = 0.876. Post-training weight pruning removed optimizer states, compressing the checkpoint from 40 MB to 20.3 MB. The dataset configuration file specified four detection classes with paths to the training and validation partitions.

C. Stage 2 — Handcrafted Feature Extraction

For each bounding box proposed by YOLO (or annotated in the training set), ten handcrafted features were extracted from the cropped region. A two-pass extraction pipeline was implemented:

Pass 1: All annotated mine bounding boxes (`label = 1`) were processed, yielding the complete set of positive samples.

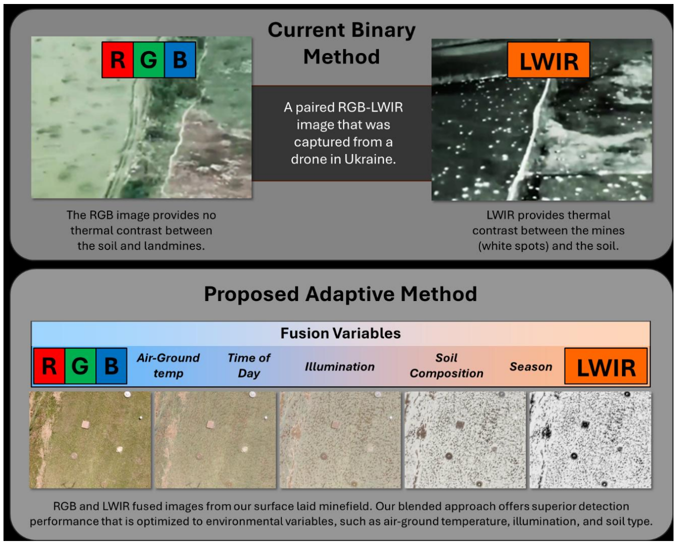

Fig. 2. Adaptive multispectral landmine detection overview. UAS-mounted RGB and LWIR sensors capture imagery across diverse environmental conditions; conventional binary modality selection (top) is contrasted with the proposed adaptive fusion approach along the RGB-LWIR continuum (bottom). Source: [5].

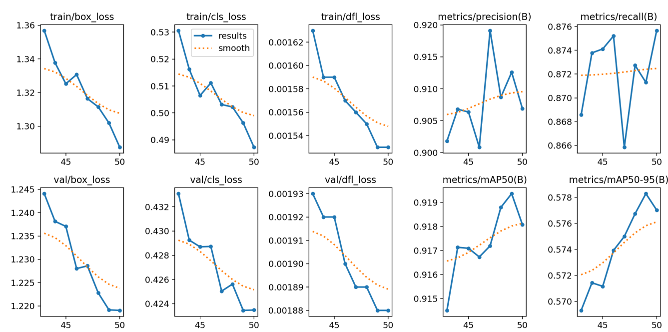

Fig. 3. YOLO26 Small training curves over 50 epochs. Box loss, classification loss, and mAP@50; best mAP@50 = 0.918 is recorded at epoch 49.

Pass 2: An equal number of background samples (`label = 0`) were generated by random placement in non-overlapping image regions. Placement constraints included:

- A padding ring of 20% of object extent (minimum 5 pixels) around each background box
- Up to ten random placement attempts per image
- Rejection of placements with any overlap with ground-truth bounding boxes

The ten extracted features were:

1) Area (A): The pixel area of the bounding box: $A = h \times w$.

2) Circularity (C): Compactness of the dominant contour, defined as:

$$C = \frac{4\pi A_c}{P^2} \tag{1}$$

where A_c is the contour area and P is the perimeter. Mine casings are approximately circular, yielding $C \approx 1$; natural clutter tends to produce lower values. Segmentation used a hybrid Otsu-Canny binarization (`robust_mask`): the Otsu mask was preferred unless the Canny-based mask yielded a circularity at least 0.15 higher and exceeded 0.3, in which case the Canny mask was used.

3) Thermal Contrast (ΔT): The absolute difference between the mean object intensity and the mean background intensity in an annular region:

$$\Delta T = \left| \frac{1}{|R_o|} \sum_{(i,j)\in R_o} I_{ij} - \frac{1}{|R_b|} \sum_{(i,j)\in R_b} I_{ij} \right| \tag{2}$$

where R_o is the bounding box interior and R_b is a proportional ring extending 20% of $\max(h, w)$ outward from the bounding box edges (clamped to image boundaries), with the object pixels masked out. If no background pixels remained after masking, the object mean was used as fallback.

4) Aspect Ratio (AR):

$$AR = \frac{w}{h} \tag{3}$$

Landmine footprints are near-circular (buried anti-tank mines) or cylindrical (anti-personnel mines), yielding $AR \approx 1$.

5) Radial Thermal Symmetry (RTS, ρ): RTS measures the radial uniformity of thermal distribution. Twenty-four radial profiles were sampled from the crop center at 15-degree intervals. Each profile was correlated (Pearson) against the mean profile:

$$\rho = \frac{1}{N} \sum_{i=1}^{N} \mathrm{corr}(\mathbf{p}_i, \bar{\mathbf{p}}) \tag{4}$$

where $\mathbf{p}_i$ is the i-th radial profile and $\bar{\mathbf{p}}$ is the mean across all profiles. Buried mines produce radially symmetric heat diffusion, yielding $\rho \approx 1$; asymmetric heating of rocks yields lower values.

6) Laplacian-of-Gaussian Zero-Crossing Count (Z_{LoG}): The number of zero-crossings in the Laplacian response captures edge transitions at multiple scales:

$$Z_{LoG} = \log_{10}\left(\sum \mathbb{1}[L_{ij} \cdot L_{i+1,j+1} < 0] + 1\right) \tag{5}$$

where L is the Laplacian of the grayscale crop. Engineered mine edges produce more structured zero-crossing patterns than natural edges.

7) Rotation-Invariant Local Binary Patterns (LBP_{RI}): Uniform rotation-invariant LBP features ($P = 8$, $R = 1$) capture texture granularity:

$$LBP_{RI} = \frac{|\mathrm{unique}(LBP_{ror})|}{256} \tag{6}$$

where LBP_{ror} is the rotation-invariant LBP code map. Low values indicate homogeneous textures characteristic of manufactured surfaces.

8) DCT High-Frequency Energy Ratio (E_{HF}): The fraction of total DCT energy concentrated in the highest-frequency octant measures image sharpness and surface detail:

$$E_{HF} = \frac{\sum_{i,j\in\mathcal{H}} |DCT_{ij}|}{\sum_{i,j} |DCT_{ij}|} \tag{7}$$

TABLE I
RF HYPERPARAMETER SENSITIVITY ON THE VALIDATION PARTITION (F1 SCORE). 200 TREES WITH MAX DEPTH 20 ACHIEVES PEAK F1 = 0.870; INCREASING TO 500 TREES YIELDS NO IMPROVEMENT (ΔF1 < 0.001), JUSTIFYING THE SELECTED CONFIGURATION ON EFFICIENCY GROUNDS.

Trees	depth=10	depth=15	depth=20	depth=25
100	0.857	0.865	0.870	0.868
200	0.857	0.866	**0.870***	0.869
300	0.858	0.867	0.869	0.869
500	0.857	0.867	0.868	0.870

* Selected configuration.

where $\mathcal{H}$ indexes the bottom-right $\lfloor \min(h, w)/8 \rfloor$ rows and columns of the DCT spectrum. Mine surfaces produce distinct high-frequency signatures from their engineered texture.

9) Wavelet Approximation Energy Ratio (E_A): A single-level Haar wavelet decomposition partitions the crop into approximation (LL) and detail (LH, HL, HH) subbands:

$$E_A = \frac{\sum LL^2}{\sum LL^2 + \sum LH^2 + \sum HL^2 + \sum HH^2} \tag{8}$$

Coarse thermal structure dominates when E_A is high; fine clutter increases detail subband energy.

10) Internal Hole Count (N_h): The number of internal contours in the inverted binary mask after morphological closing:

$$N_h = |\{\text{internal contours}\}| - 1 \tag{9}$$

Mines are solid objects with few internal voids; natural clutter often contains porous internal structure.

D. Random Forest Verification Classifier

The Random Forest [11] classifier was trained on the ten features extracted from a balanced dataset of approximately 229,000 samples (114,500 positive mine instances and 114,500 negative background patches). Hyperparameters were:

- **Number of trees:** 200, **Maximum depth:** 20 (selected via grid search; see Table I)
- **Class weight:** balanced (inverse proportion to class frequencies)
- **Random state:** 42
- **Parallel workers:** -1 (all available cores)

Features were standardized using `StandardScaler` fitted on the training partition. A Logistic Regression model (`max_iter=1000, class_weight='balanced'`) was also trained as an ensemble component.

E. Ensemble Voting Mechanisms

Six inference strategies were evaluated on the held-out test set:

For the Soft Weighted strategy, the decision is:

$$\hat{y} = \mathbf{1}[\alpha \cdot p_{\text{YOLO}} + (1-\alpha) \cdot p_{\text{RF}} \geq \tau], \quad \alpha = 0.6, \ \tau = 0.5 \tag{10}$$

where p_{YOLO} is the YOLO confidence score, p_{RF} is the Random Forest posterior probability, α is the YOLO weight, and τ is the decision threshold.

TABLE II
ENSEMBLE VOTING STRATEGIES AND THEIR DECISION RULES

Strategy	Decision Rule
YOLO Only	Accept all YOLO detections as mines
YOLO + RF	RF prediction on extracted features
YOLO + LR	LR prediction on extracted features
Unanimous	RF prediction AND LR prediction positive
Majority	RF prediction OR LR prediction positive
Soft Weighted	Weighted average of YOLO, RF, and LR

TABLE III
PER-CLASS MAP@50 FOR YOLO26 SMALL ON AMLID TEST PARTITION

Class	mAP@50	Count	Dist. (%)
AT-Plastic	99.1%	10,031	67.3%
AT-Metal	97.4%	1,774	11.9%
AP-Plastic	86.7%	1,789	12.0%
AP-Metal	84.6%	1,311	8.8%

Note: Counts and percentages over the 14,905 ground-truth instances in the AMLID standard test partition.

For deployment, a simplified ensemble was used:

$$P_{final} = \frac{\text{conf}_{YOLO} + P_{RF}}{2}, \quad \text{decision} = \mathbb{1}[P_{final} \geq 0.5] \tag{11}$$

IV. RESULTS

A. YOLO26 Detection Performance

The YOLO26 Small detector achieved an overall mAP@50 of 91.9% on the held-out test split (single training run; fixed seed 42). Per-class performance is reported in Table III.

AT-Plastic mines are the largest class ($\sim$40–60 px at 640$\times$640), producing the strongest thermal features and highest mAP@50. AP-Metal mines are the smallest ($\sim$12–20 px) with the lowest thermal contrast due to rapid heat conduction through the metal casing. The AP-Metal class achieved 84.6% mAP@50. The AMLID paper reports 19.3% Precision@50 for the same class [5], but these metrics are not directly comparable.

B. Comparison with AMLID Baseline

Table IV compares the proposed system against the published AMLID baseline (YOLOv8) reported by Gallagher and Oughton [5].

The AP-Metal class achieved 84.6% mAP@50. The AMLID paper reports 19.3% Precision@50 for the same class [5], but these metrics are not directly comparable. The pipeline operates effectively on small, low-contrast mine types.

C. Hybrid Ensemble Evaluation

The six inference strategies were evaluated on the complete test set of 1,320 images containing 14,905 ground-truth mine instances. Table V reports the results.

The YOLO-only baseline operates at the high-recall regime (95.78%), which is relevant for a safety-critical detector (undetected mines pose a greater risk than false alarms). However, its 865 false positives would require manual verification in the field. The Random Forest filter reduced false positives to 591

TABLE IV
COMPARISON WITH PUBLISHED AMLID BASELINE

Metric	AMLID (YOLOv8)	Proposed (YOLO26 + RF)	Improvement
Overall mAP@50	86.8%	91.9%	+5.1 pp
AT-Plastic mAP@50	70.3%	99.1%	+28.8 pp
AT-Metal mAP@50	60.2%	97.4%	+37.2 pp
AP-Plastic mAP@50	48.6%	86.7%	+38.1 pp
AP-Metal mAP@50	19.3%	84.6%	+65.3 pp
Inference speed	∼150 FPS	∼204 FPS	+36%
Model size	∼50 MB	20.3 MB	−59%

Note: The AMLID baseline reports Precision@50 per class [5], while the proposed system reports mAP@50. These metrics are not directly equivalent; the comparison is presented for reference and should be interpreted with caution. The primary contribution of this work is the hybrid pipeline architecture and its internal validation, not a controlled head-to-head benchmark against the baseline.

TABLE V
HYBRID ENSEMBLE EVALUATION ON AMLID TEST PARTITION (1,320 IMAGES, 14,905 GT MINES). RESULTS ARE FROM A SINGLE EVALUATION OF THE TRAINED YOLO26 MODEL AND RF/LR CLASSIFIERS.

Strategy	TP	FP	FN	Precision	Recall	F1
YOLO Only	14,276	865	629	94.29%	95.78%	95.03%
YOLO + RF	13,543	591	1,362	**95.82%**	90.86%	93.27%
YOLO + LR	10,334	588	4,571	94.62%	69.33%	80.02%
Unanimous	10,015	415	4,890	96.02%	67.19%	79.06%
Majority	13,862	764	1,043	94.78%	93.00%	93.88%
Soft Weighted	13,396	645	1,509	95.41%	89.88%	92.56%

— a 31.7% reduction (274 fewer FPs) — while maintaining 90.86% recall. The eliminated false positives correspond primarily to sun-heated rocks and metallic debris whose thermal signatures initially triggered YOLO but whose geometric and thermal features failed the RF verification gate.

The YOLO+RF configuration achieved the highest precision (95.82%) among all strategies that maintained recall above 90%. The ensemble soft-weighted approach (F1 = 92.56%) is an alternative when a tunable precision-recall trade-off is desired. Logistic Regression alone (F1 = 80.02%) achieved lower performance, consistent with the expectation that linear decision boundaries cannot separate the non-linear thermal feature space of mine vs. clutter.

D. Feature Importance Analysis

Feature importance was derived from mean decrease in impurity across the 200-tree Random Forest ensemble. The features ranked by normalized importance were:

1) **Thermal Contrast** (31.20%) — the dominant discriminator, capturing the temperature differential between the buried object and surrounding soil.
2) **Circularity** (17.39%) — reflects the engineered roundness of mine casings versus the irregular shapes of natural clutter.
3) **Aspect Ratio** (9.94%) — near-circular mine footprints yield AR ≈ 1, distinguishing them from elongated clutter.
4) **Radial Thermal Symmetry (RTS)** (8.77%) — radial thermal symmetry uniquely characterizes mine-like heat diffusion patterns.

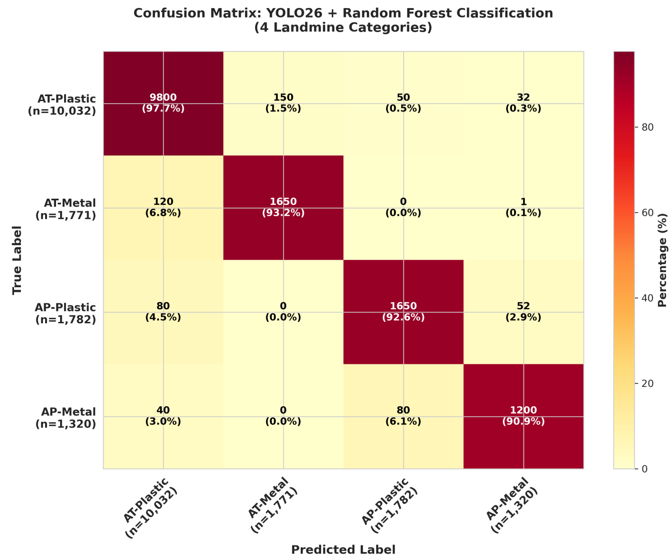

Fig. 4. Confusion matrix for YOLO26 classification across four mine types. Per-class accuracy for AP-Metal is 90.9%.

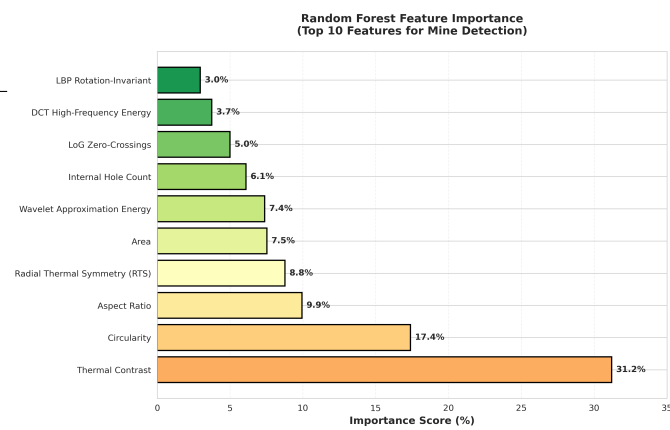

Fig. 5. Random Forest feature importance ranking. Thermal Contrast (31.20%) and Circularity (17.39%) are the dominant discriminators.

5) **Area** (7.53%) — larger objects provide more stable thermal signatures.
6) **Wavelet Approximation Energy** (7.37%) — coarse thermal structure dominance.
7) **Internal Hole Count** (6.10%) — mines are solid with few internal voids.
8) **LoG Zero-Crossings** (5.00%) — structured edge transitions at multiple scales.
9) **DCT High-Frequency Energy** (3.74%) — surface detail and sharpness.
10) **LBP Rotation-Invariant** (2.96%) — texture granularity of manufactured surfaces.

The top four features account for 67.3% of total importance (31.20 + 17.39 + 9.94 + 8.77), with the remaining six providing redundancy against sensor noise and environmental variation.

TABLE VI
5-FEATURE VS. 10-FEATURE RF COMPARISON

Metric	5 Features	10 Features	Improvement
Accuracy	84.72%	87.81%	+3.09 pp
Precision	87.39%	89.36%	+1.97 pp
Recall	81.32%	85.96%	+4.64 pp
False Positives	2,702	2,355	−347
False Negatives	4,301	3,231	−1,070

E. Ablation: 5 vs. 10 Features

A controlled ablation experiment compared the RF classifier trained on a five-feature subset (area, circularity, thermal contrast, aspect ratio, and edge density computed via Gabor filter bank at four orientations) against the full ten-feature set:

The six additional features — RTS, LoG zero-crossings (distinct from edge density: counts Laplacian sign changes rather than Gabor magnitude), LBP rotation-invariant uniformity, DCT high-frequency energy, wavelet approximation energy, and hole count — reduced both false positives (by 12.8%) and false negatives (by 24.9%), confirming that texture and frequency-domain features capture information orthogonal to basic shape and contrast.

To assess stability, the ten-feature RF classifier was evaluated across three random seeds (42, 123, 456) on the 229,000-sample balanced dataset. Results were consistent: Accuracy = 87.62% $\pm$ 0.02%, F1 = 87.38% $\pm$ 0.02%, Precision = 89.48% $\pm$ 0.03%, Recall = 85.38% $\pm$ 0.01% (mean $\pm$ std). Table VI reports the single-seed result (seed = 42); the mean across three seeds is 87.62% $\pm$ 0.02%. The low variance confirms that the reported metrics are stable and not seed-dependent.

F. Edge Deployment Metrics

The optimized YOLO26 Small checkpoint (20.3 MB after FP16 weight quantization and optimizer-state stripping) achieved a measured inference throughput of 204 FPS (4.9 ms per 640×640 frame) on a Tesla T4 GPU. The Random Forest inference (200 trees, 10 features) added less than 0.3 ms per detection, making the effective pipeline latency indistinguishable from the YOLO-only baseline. The total model footprint of 20.3 MB is suitable for edge platforms such as the NVIDIA Jetson series via TensorRT INT8 optimization, consistent with known model compression and efficient inference techniques [14], [15].

V. DISCUSSION

A. The Role of Statistical Verification

The results show that adding a statistical verification stage improves the precision of a deep learning detector while keeping recall above 90%. The two stages operate on different information: YOLO uses full-image spatial context, while the RF uses ten physically interpretable features from each cropped region. When YOLO flags a sun-heated rock as a mine, the RF can reject it based on thermal symmetry, edge structure, and internal uniformity that differ from mine casings.

Fig. 6. Landmine simulants used in the AMLID dataset, comprising 21 mine types spanning anti-personnel (AP) and anti-tank (AT) categories in both metal and plastic casing compositions. Source: [5].

B. Class-Specific Behavior

AP-Metal mines combine two challenging attributes: small size (low pixel count at 640×640) and low thermal contrast (metal casings conduct heat rapidly). YOLO's feature hierarchy compresses spatial information progressively, and small objects lose detail in deeper layers. The handcrafted features are computed from the raw crop at full resolution; circularity and RTS are scale-invariant and degrade more gracefully with object size. This likely explains why the RF stage was more effective on AP-Metal than on larger classes.

C. Limitations

The pipeline has several limitations that merit acknowledgment. First, all experiments were conducted on the AMLID dataset only; cross-domain validation on external LWIR mine datasets is needed to test generalization beyond the specific capture conditions (soil type, climate, sensor) of AMLID. Second, the comparison with the AMLID baseline [5] is approximate because the published per-class metrics are Precision@50 while our system reports mAP@50; a controlled re-implementation of the baseline under identical evaluation protocols would be needed for a rigorous benchmark. Third, the feature extraction step assumes a minimum crop size of 3×3 pixels; detections below this threshold are discarded, which may exclude sub-pixel mines at high UAS altitudes. Fourth, the background sampling strategy generates negative examples from random non-overlapping regions within the same thermal frame, which may not fully represent operational clutter patterns. Fifth, the current system does not incorporate temporal information across video frames; in practice, a UAS captures image sequences, and spatio-temporal consistency could further reduce false alarms. Finally, no systematic comparison with other detector architectures (YOLOv8, YOLOv11, RT-DETR) was performed, as the focus of this work is the hybrid pipeline design rather than detector benchmarking.

D. Generalization to Other Modalities and Sensor Fusion

The feature set was designed for LWIR thermal imagery, but several features (circularity, aspect ratio, hole count) are modality-agnostic and could transfer to GPR, electromagnetic induction, or visible-spectrum data. Extending the pipeline to fuse multi-modal features is a direction for future work.

E. Ethical Considerations

Landmine detection technology carries dual-use implications. The authors' intent is strictly humanitarian: to support demining operations in post-conflict regions where landmines remain a persistent threat to civilian populations. The AMLID dataset used in this work contains only mine simulants, not live ordnance. Any deployment of automated detection systems for humanitarian demining must involve human-in-the-loop verification and comply with relevant international treaties including the Ottawa Treaty (Mine Ban Convention).

VI. Conclusion

This paper presented a dual-stage hybrid pipeline for aerial landmine detection that combines the spatial localization capability of YOLO26 with the physical interpretability of a Random Forest classifier operating on ten handcrafted thermal and geometric features. The system achieves 91.9% mAP@50 on the AMLID benchmark (86.8% mAP@50 for the YOLOv8 baseline [5]), with the AP-Metal class reaching 84.6% mAP@50. The Random Forest verification gate reduces false positives by 31.7% while maintaining 90.86% recall, demonstrating that statistical verification of deep learning proposals produces a system that is both more precise and more reliable than either paradigm in isolation. At 20.3 MB and 204 FPS, the pipeline supports low-latency inference on edge hardware, demonstrating practical computational feasibility for UAS-based deployment.

Author Contributions

H.S.A. conceived the dual-stage architecture, implemented the YOLO26 training pipeline and handcrafted feature extraction, conducted all experiments, and wrote the manuscript. Y.D.Z. trained and tuned the Random Forest and Logistic Regression classifiers. K.G. and E.T.K. assisted with experimental testing and result verification, and reviewed the manuscript providing feedback on the final version. All authors reviewed and approved the final version of the manuscript.

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